ORANGE MANUAL

FUNDAMENTAL OF AI & ML

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1. ABSTRACT

Orange Data Mining is an open-source machine learning and data visualization software designed to make data analysis simple and interactive. It provides a graphical, drag-and-drop interface that allows users to build machine learning workflows without writing complex code. Orange is built using Python and supports a wide range of data mining techniques such as classification, regression, clustering, feature selection, and model evaluation.

The main purpose of Orange is to simplify the machine learning process for students, researchers, and professionals. It is widely used in academic institutions for teaching data science concepts because it allows users to visually understand how algorithms work and how data flows through different stages of processing.

2. INTRODUCTION TO ORANGE

Orange was developed by the Bioinformatics Laboratory at the University of Ljubljana. It is designed as a visual data mining tool where users connect different components, known as widgets, to create a workflow. Each widget performs a specific task such as loading data, preprocessing, training a model, or evaluating performance.

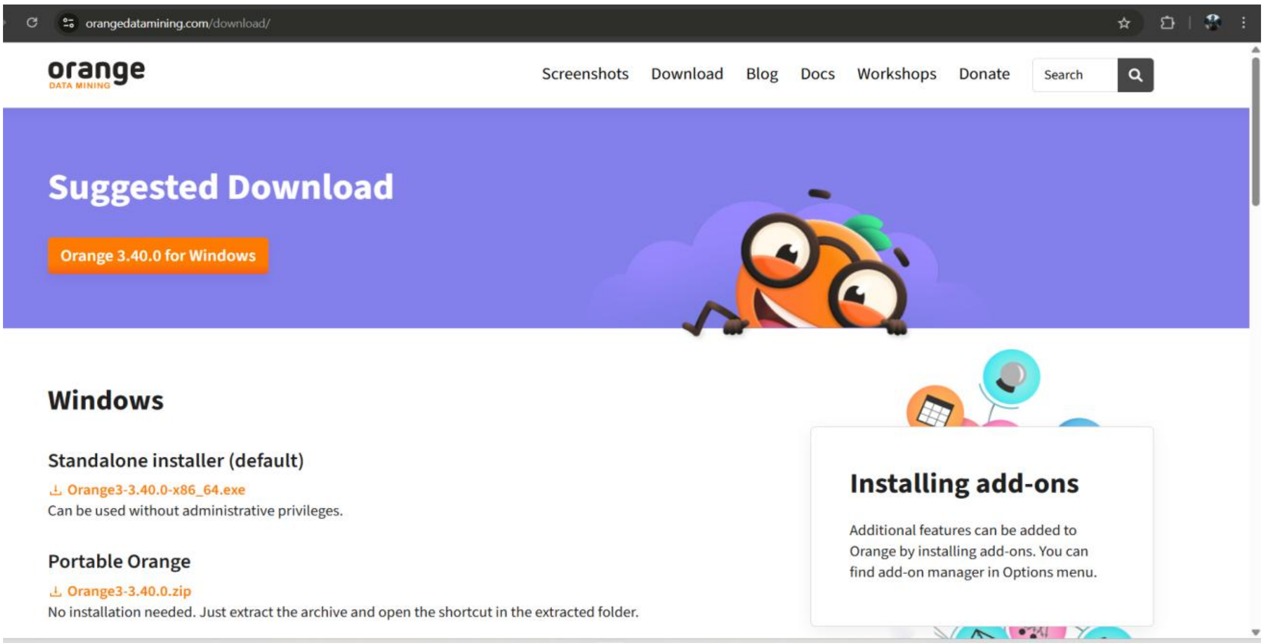
In real-world machine learning applications, raw data is rarely perfect. It may contain missing values, incorrect formatting, or irrelevant features. Before applying any algorithm, the data must be cleaned and structured properly. Orange provides powerful preprocessing tools that make this process simple and systematic.

The visual workflow design helps users understand each step clearly, making it easier to debug and improve models.

3. INSTALLATION OF ORANGE

Installing Orange is straightforward. Below is the detailed process:

1. Open any web browser such as Chrome or Edge.
2. Search for “Orange Data Mining official website”.
3. Open the [official website.]



4. Click on the “Download” option.
5. Select the Windows version.
6. Download the installer (.exe file).
7. Once downloaded, double-click the file.
8. If a security pop-up appears, click “Yes”.
9. The setup wizard will open. Click “Next”.
10. Read and accept the license agreement.
11. Choose the installation location (default is recommended).
12. Click “Install”.
13. Wait for the installation to complete.
14. Click “Finish”.

After installation, an Orange icon will appear on the desktop. Double-click it to launch the software.

4. UNDERSTANDING THE ORANGE INTERFACE

When **Orange Data Mining** is launched, the main working window that appears is called the **Canvas**. The Canvas serves as the central workspace where users design, build, and manage their machine learning workflows visually. Unlike traditional programming environments where users write code, Orange allows users to create workflows by simply dragging and connecting widgets. This visual approach makes the platform highly beginner-friendly and easy to understand.

On the left-hand side of the interface, there is a panel containing various widgets. These widgets are organized into categories based on their function. The most commonly used categories include:

- **Data** – for loading and managing datasets
- **Visualize** – for creating graphs and visual analysis
- **Model** – for applying machine learning algorithms
- **Evaluate** – for testing and measuring model performance
- **Transform** – for preprocessing and modifying data
- **Unsupervised** – for clustering and pattern discovery

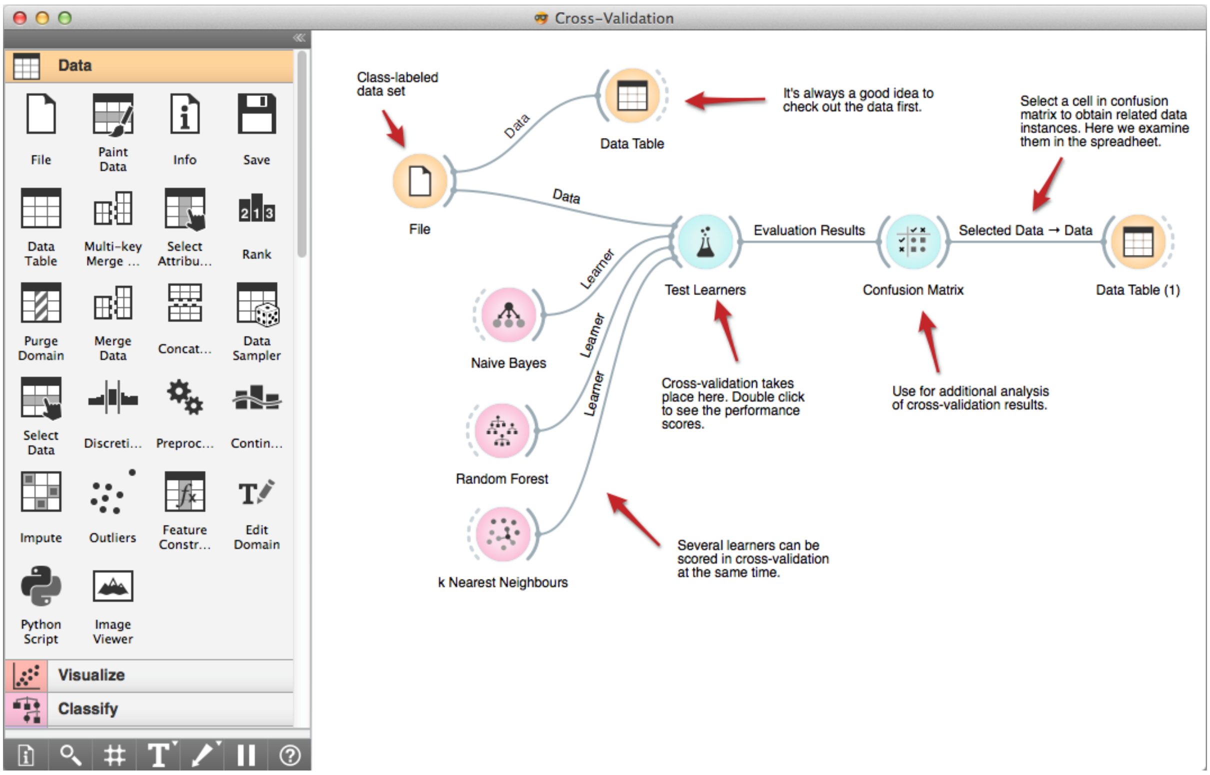
Each widget performs a specific task in the machine learning process. For example, the File widget loads data, while the Test & Score widget evaluates model accuracy.

The center of the screen is the workflow area (Canvas) where widgets are dragged and connected. Widgets are connected using lines, which represent the flow of data from one step to another. The output of one widget becomes the input of the next, forming a logical pipeline.

At the top of the interface, the **Menu Bar** contains options such as:

- File (create, open, save workflows)
- Options (system settings)
- View (adjust layout and visibility)
- Help (documentation and support)

This entire structure makes Orange highly interactive and suitable for academic and research purposes.



5. LOADING A DATASET

The first and most essential step in any machine learning project is loading the dataset. In Orange, this is done using the **File widget**, located under the Data category.

To load a dataset:

1. Drag the **File widget** onto the Canvas.
2. Double-click it to open its settings.
3. Browse and select a dataset from your system.

Orange supports multiple file formats including:

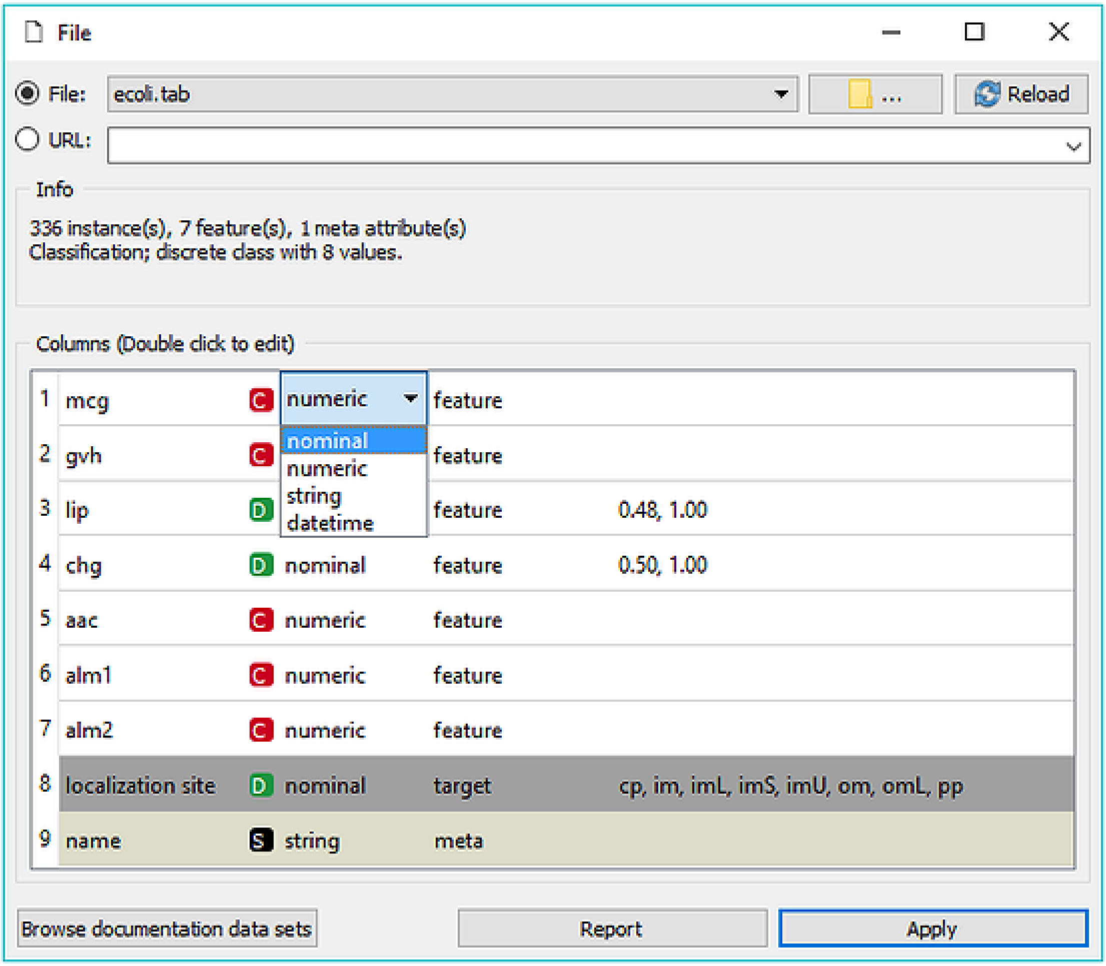
- CSV (Comma Separated Values)
- Excel (.xlsx)
- ARFF (Attribute-Relation File Format)
- Tab-delimited files

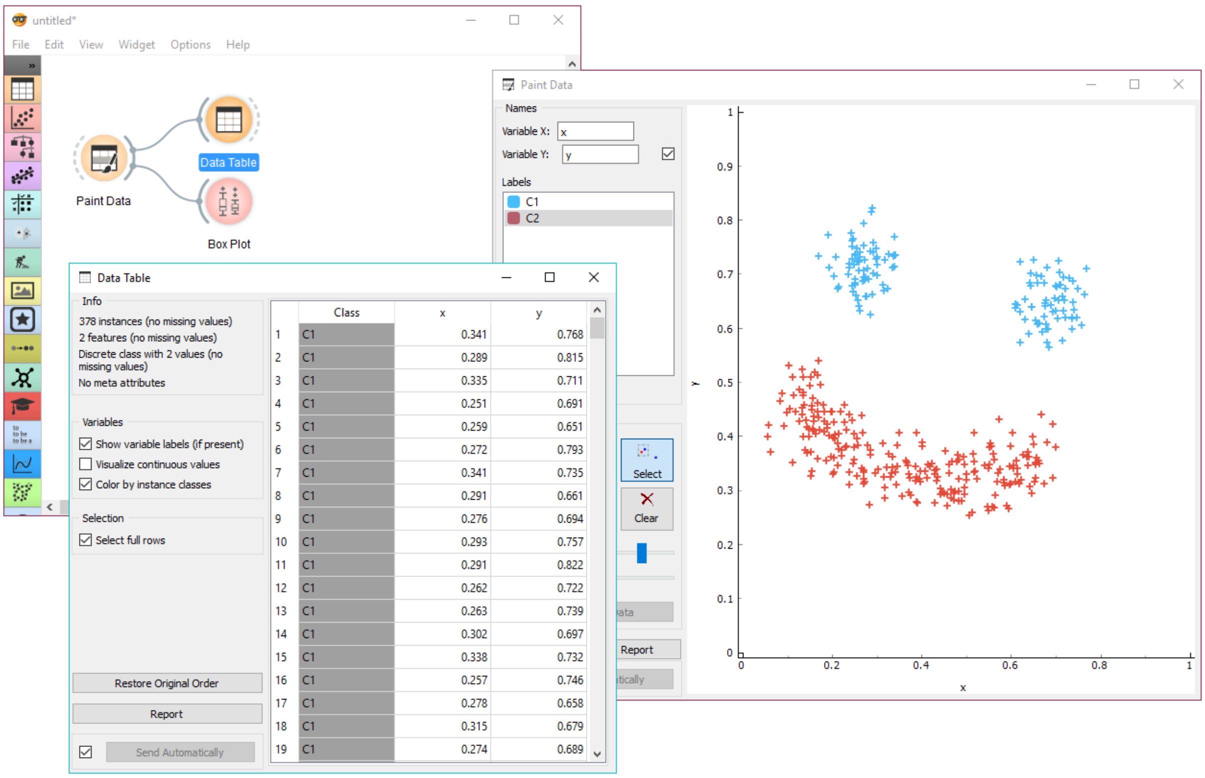
After selecting the dataset, Orange automatically reads the file and displays:

- Number of instances (rows)
- Number of features (columns)
- Type of each variable (numeric, categorical, text, etc.)
- Target/class variable (if defined)

To view the dataset in a structured table format, the **Data Table widget** can be connected to the File widget. This allows users to inspect the raw data, check for missing values, and understand how attributes are organized before applying any model.

Understanding the dataset structure at this stage is very important because incorrect attribute types or wrongly assigned class variables can affect model results.


File
File: ecoli.tab
Reload
URL:
Info
336 instance(s), 7 feature(s), 1 meta attribute(s)
Classification; discrete class with 8 values.
Columns (Double click to edit)
1 mcg numeric feature
nominal
numeric
string
datetime
2 gvh feature
3 lip feature 0.48, 1.00
4 chg nominal feature 0.50, 1.00
5 aac numeric feature
6 alm1 numeric feature
7 alm2 numeric feature
8 localization site nominal target cp, im, imL, imS, imU, om, omL, pp
9 name string meta
Browse documentation data sets
Report
Apply


untitled*
File Edit View Widget Options Help
Paint Data
Data Table
Box Plot
Paint Data
Names
Variable X: x
Variable Y: y
Labels
C1
C2
Select
Clear
Report
Data Table
Info
378 instances (no missing values)
2 features (no missing values)
Discrete class with 2 values (no missing values)
No meta attributes
Variables
Show variable labels (if present)
Visualize continuous values
Color by instance classes
Selection
Select full rows
Restore Original Order
Report
Send Automatically
Class
x
y

6. DATA PREPROCESSING

Data preprocessing is a critical stage because real-world datasets are rarely clean. They may contain:

- Missing values
- Duplicate records
- Irrelevant features
- Outliers
- Inconsistent formatting

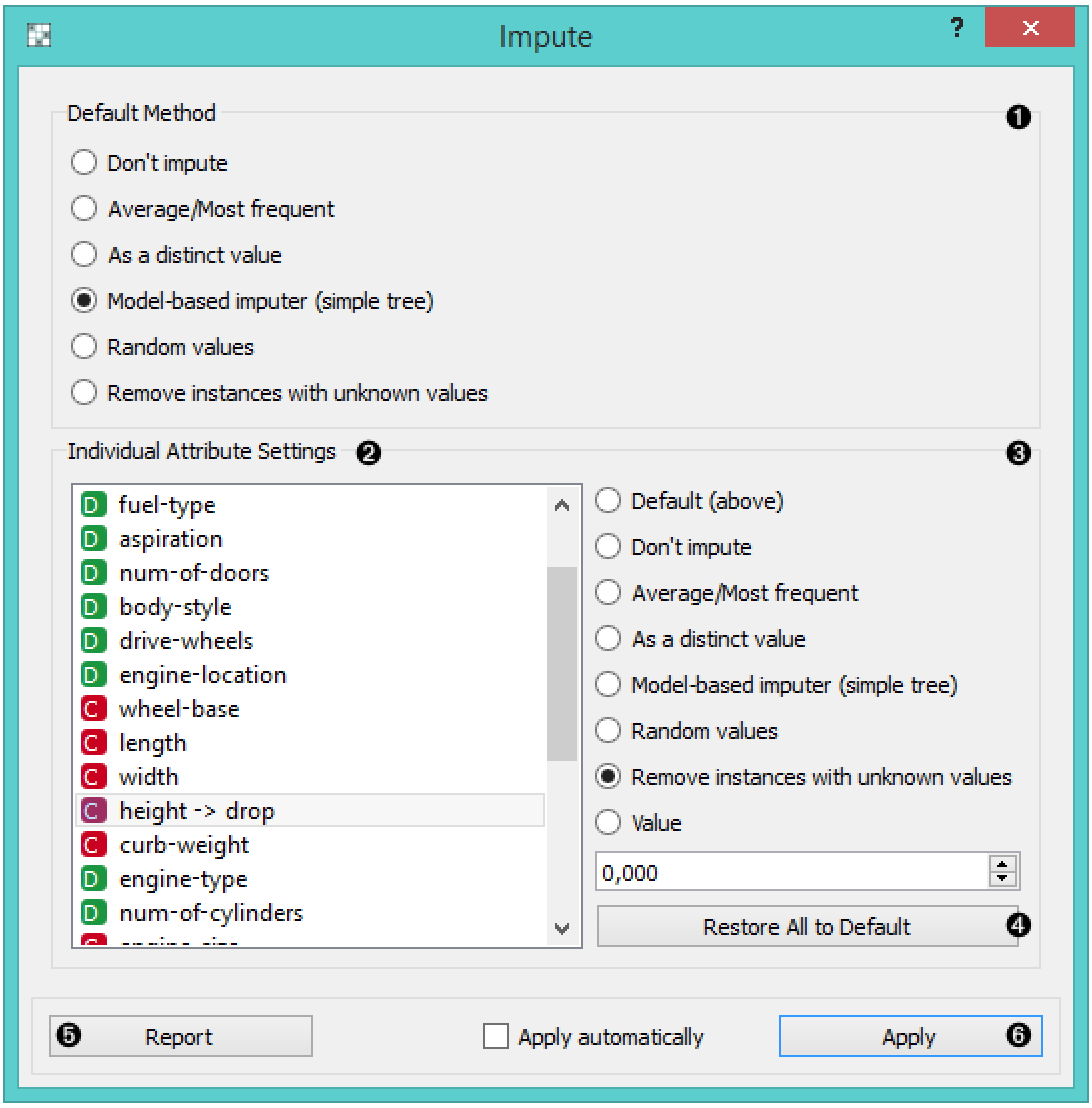
If raw data is directly used for training, the machine learning model may produce inaccurate results. Therefore, Orange provides multiple widgets for cleaning and preparing data.

Handling Missing Values – Impute Widget

The **Impute widget** is used to replace missing values. After connecting it to the File widget:

- Numeric values can be replaced with mean or median.
- Categorical values can be replaced with the most frequent category.
- Custom replacement strategies can also be applied.

This ensures that the dataset contains no null entries that might cause errors during training.



Feature Selection – Select Columns Widget

The **Select Columns widget** allows users to:

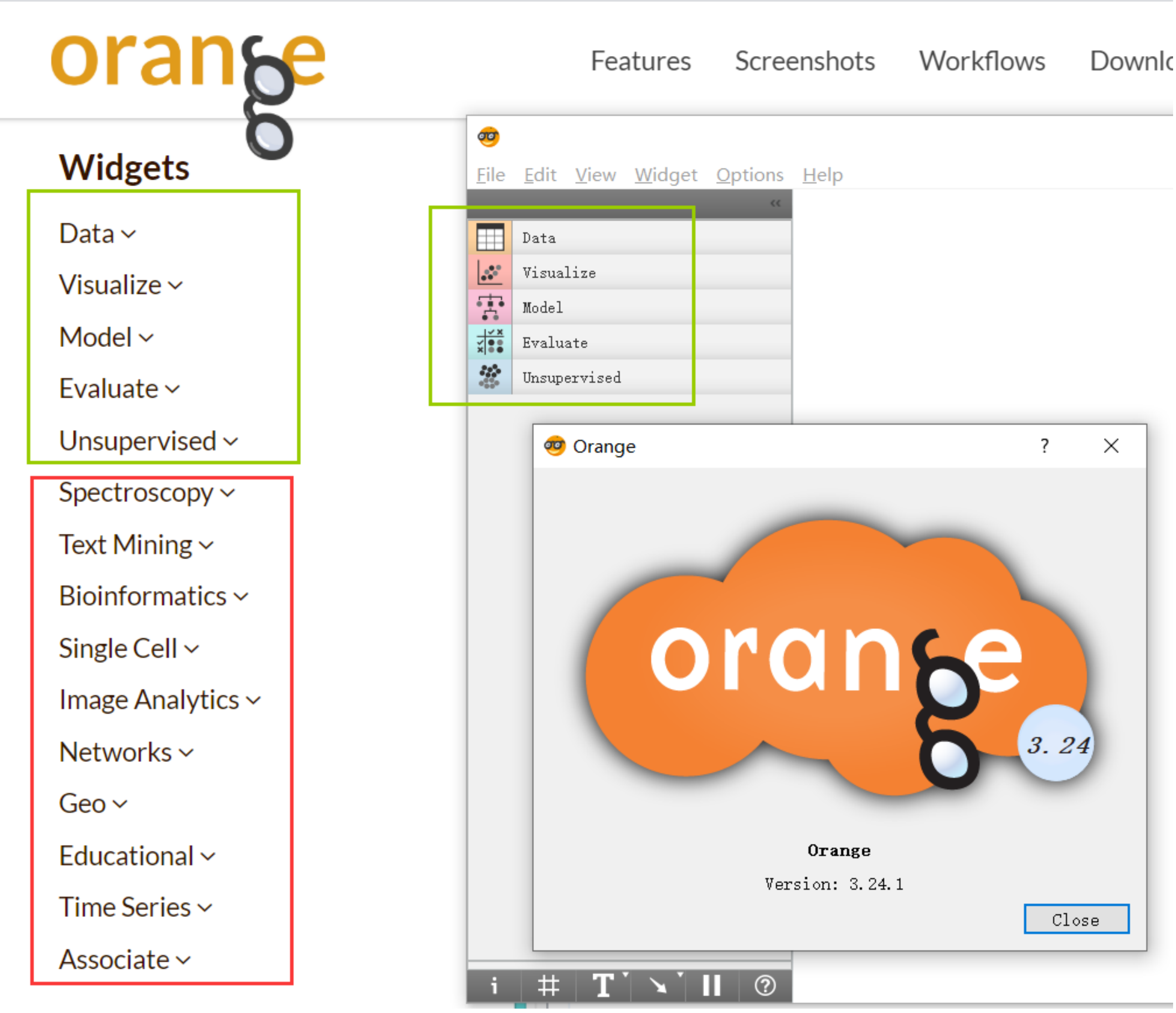
- Remove irrelevant attributes
- Assign a class (target) variable
- Define meta attributes

Removing unnecessary features reduces noise and improves model efficiency. It also helps in reducing overfitting.

Normalization – Normalize Widget

Some algorithms, especially **k-Nearest Neighbors (kNN)** and distance-based models, are sensitive to scale differences. If one attribute has values in thousands and another in decimals, the larger one may dominate calculations.

The **Normalize widget** scales numeric attributes into a standard range (such as 0–1). This ensures fair comparison among features.



7. CLASSIFICATION IN ORANGE

Classification is a supervised learning method used when a dataset contains a predefined class label. The objective is to predict the correct class for new data instances.

In Orange, classification begins by defining the class variable in the File or Select Columns widget.

Common classification algorithms available in Orange include:

- Logistic Regression
- Decision Tree
- Random Forest
- Naive Bayes
- k-Nearest Neighbors

After connecting a model to the dataset, it must be evaluated. This is done using the **Test & Score widget**.

Model Evaluation

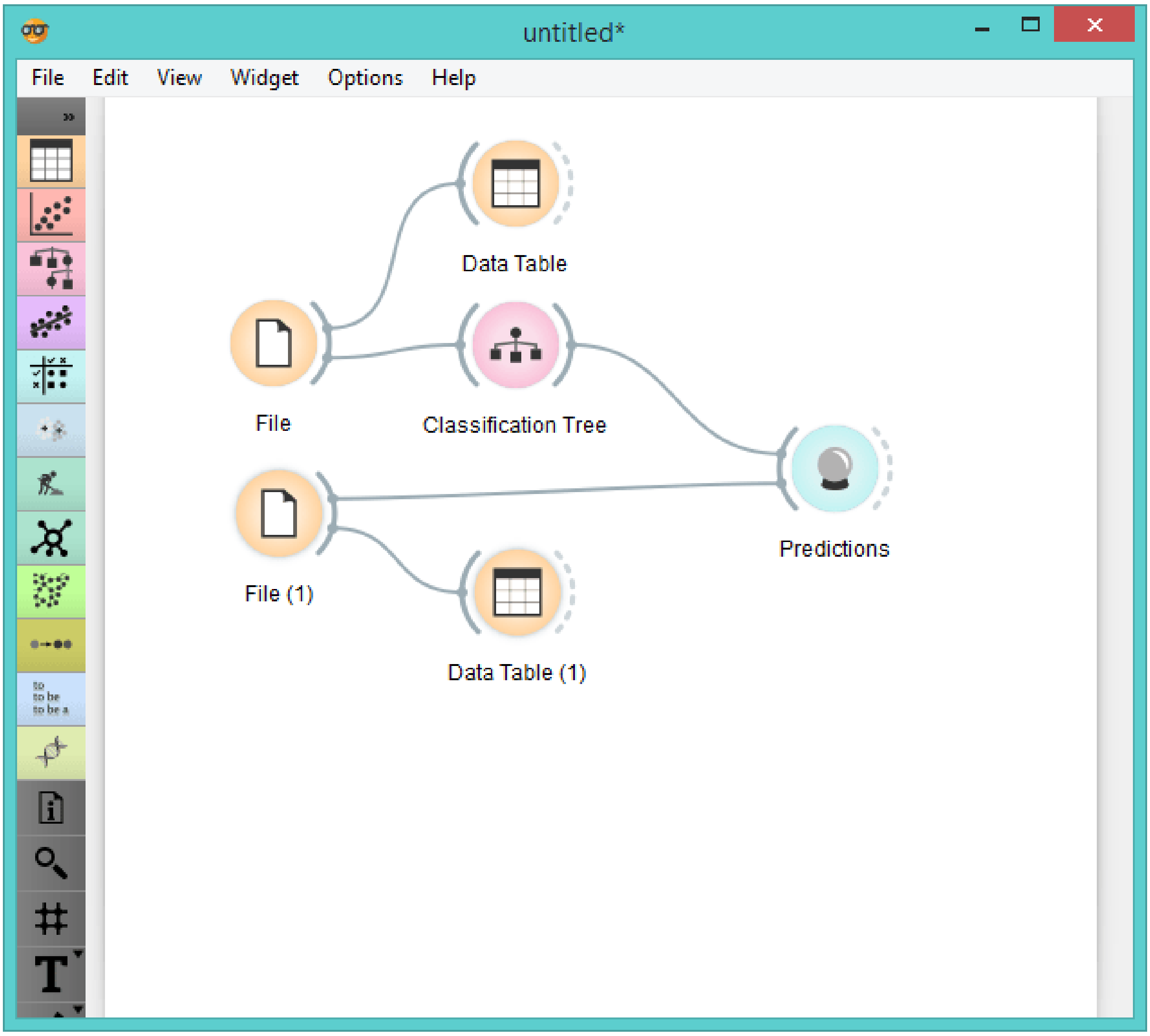
The Test & Score widget provides evaluation methods such as:

- Cross-validation (recommended)
- Percentage split
- Leave-one-out

It calculates performance metrics including:

- Accuracy
- Precision
- Recall
- F1 Score
- AUC (Area Under Curve)

Cross-validation is preferred because it divides the dataset into multiple parts and tests the model several times, giving more reliable results.

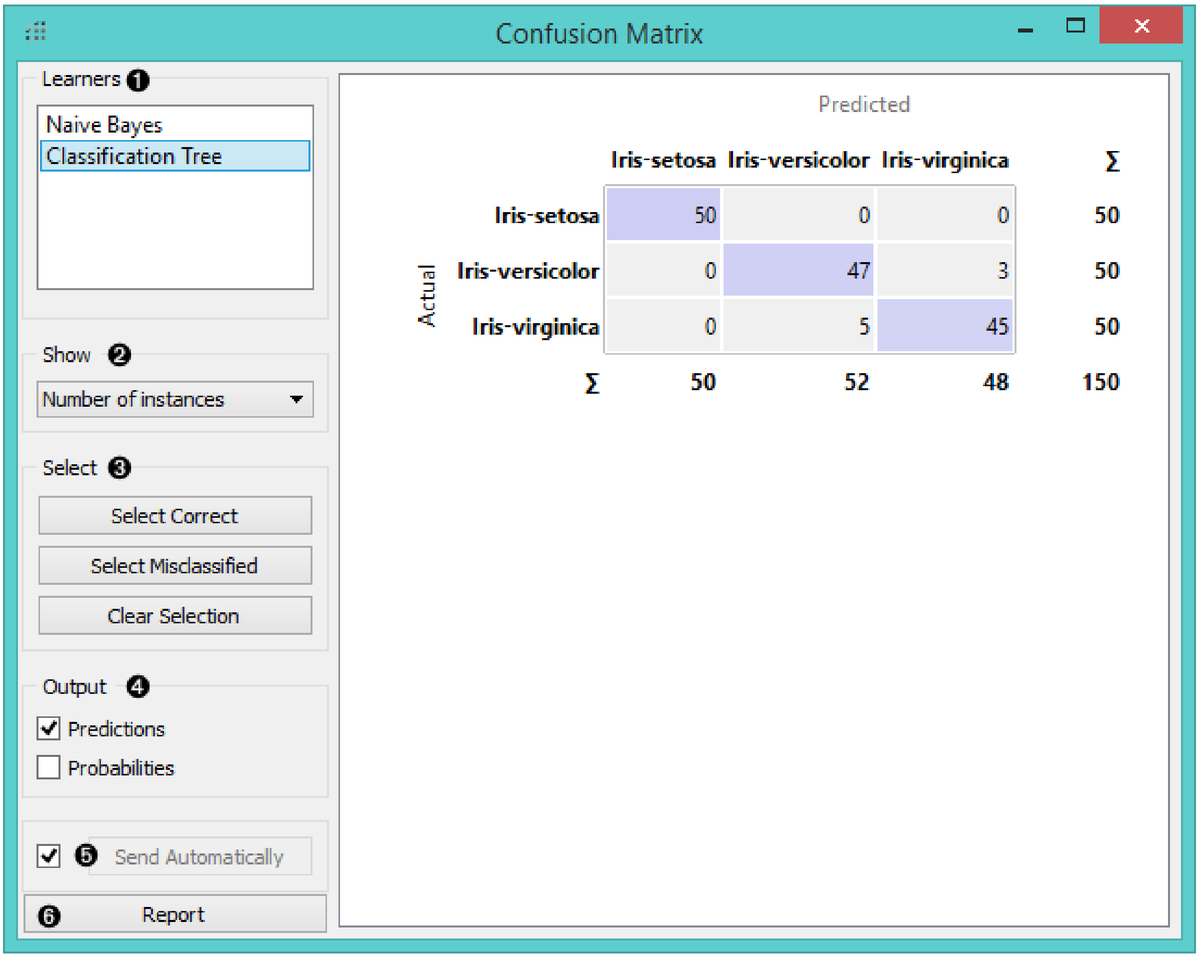


Confusion Matrix

The **Confusion Matrix widget** displays:

- True Positives
- True Negatives
- False Positives
- False Negatives

This helps users understand which class is being misclassified and analyze model weaknesses.


Confusion Matrix
Learners 1
Naive Bayes
Classification Tree
Show 2
Number of instances
Select 3
Select Correct
Select Misclassified
Clear Selection
Output 4
Predictions
Probabilities
5 Send Automatically
6 Report
Predicted
Iris-setosa Iris-versicolor Iris-virginica ∑
Actual
Iris-setosa 50 0 0 50
Iris-versicolor 0 47 3 50
Iris-virginica 0 5 45 50
∑ 50 52 48 150

8. CLUSTERING IN ORANGE

Clustering is an unsupervised learning technique used when there is no predefined class variable. It groups similar data points based on patterns and similarity.

In Orange, clustering algorithms include:

- K-Means
- Hierarchical Clustering
- DBSCAN

After connecting a clustering algorithm to the dataset:

- The number of clusters can be specified (in K-Means).
- The model automatically groups similar instances.

The **Scatter Plot widget** can then be connected to visualize clusters. Different colors represent different groups, making pattern identification easier.

9. PREDICTION USING TRAINED MODEL

Once a model has been trained and evaluated, it can be used to predict outcomes for new unseen data.

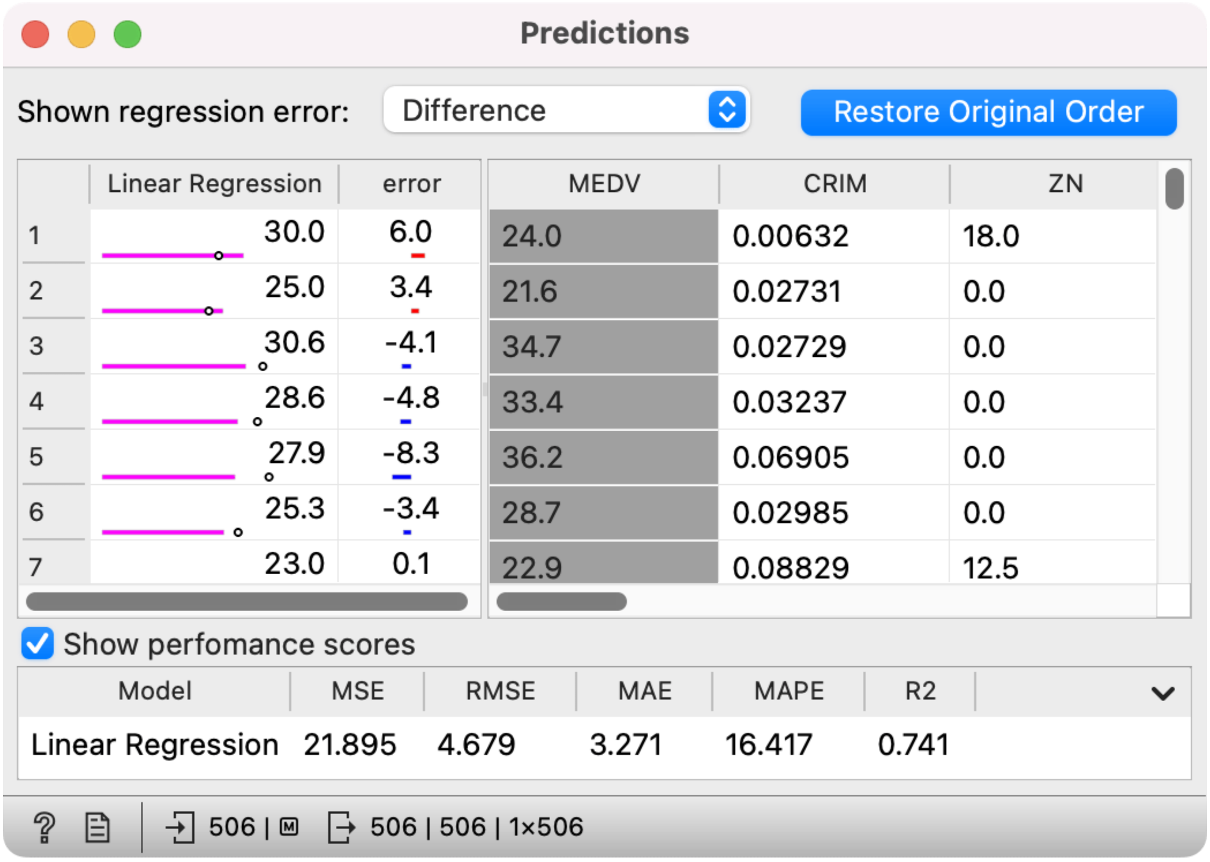
Steps:

1. Load new data using another File widet.
2. Connect the trained model to the **Predictions widget**.
3. Connect the new dataset to the Predictions widget.

The output includes:

- Predicted class label
- Probability score for each class

This is useful in real-world applications such as spam detection, disease prediction, or customer classification.



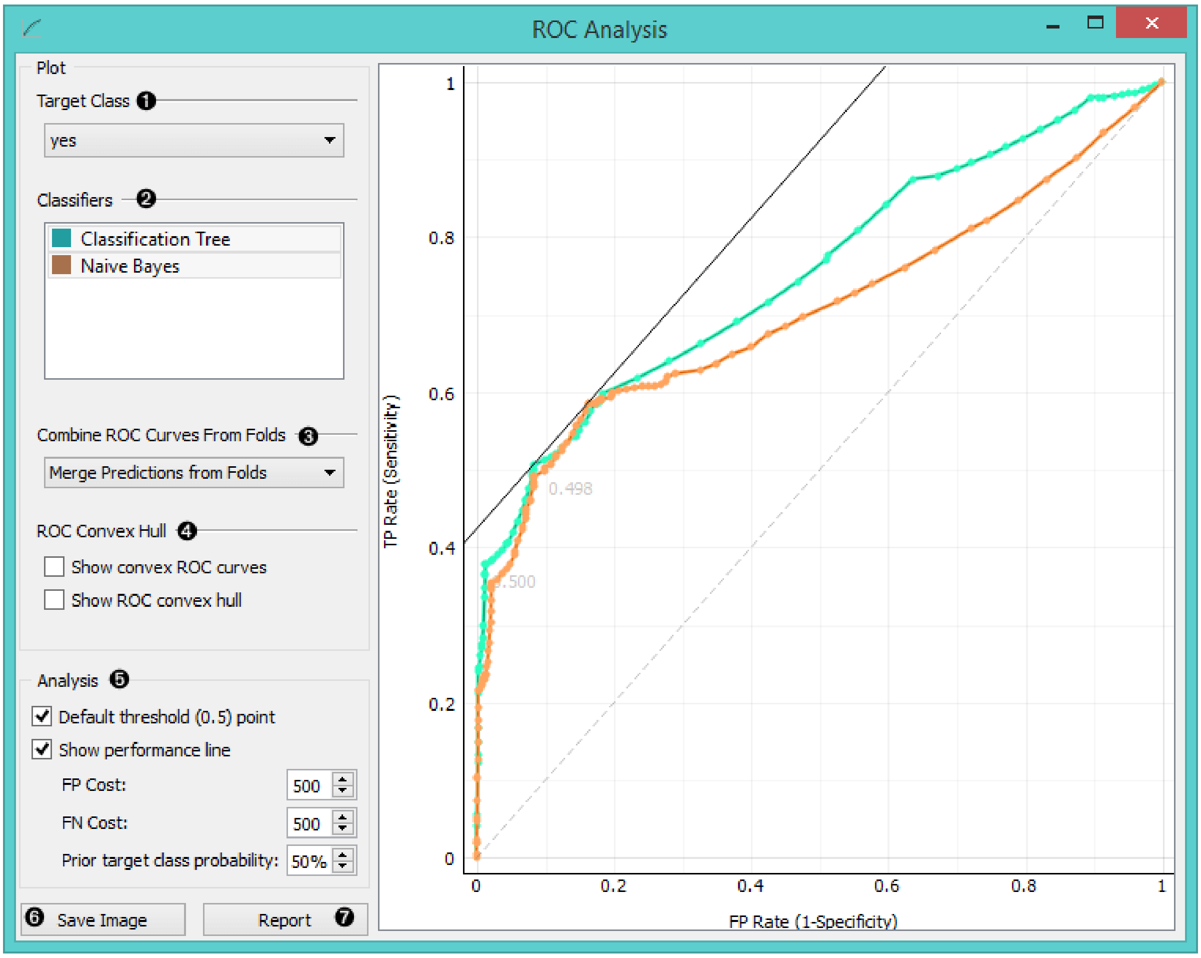
10. VISUALIZATION TOOLS

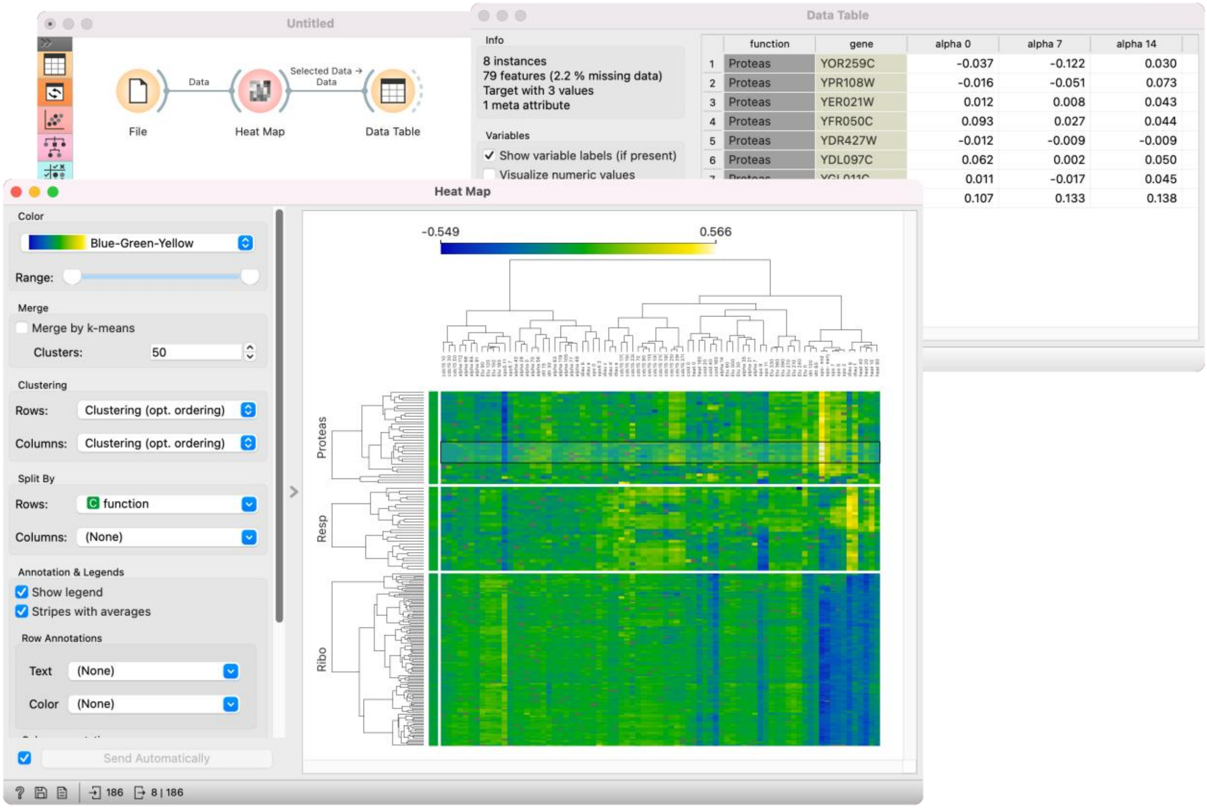
Visualization is one of the strongest features of Orange. It provides interactive and dynamic graphs that make analysis intuitive.

Important visualization tools include:

- **Scatter Plot** – compares two attributes
- **Box Plot** – shows distribution and outliers
- **Heat Map** – displays correlations
- **ROC Analysis** – compares classification performance

The **ROC Curve** visually compares classifiers by plotting True Positive Rate vs False Positive Rate.




Untitled
Data
Selected Data → Data
File
Heat Map
Data Table
Info
8 instances
79 features (2.2 % missing data)
Target with 3 values
1 meta attribute
Variables
Show variable labels (if present)
Visualize numeric values
function
gene
alpha 0
alpha 7
alpha 14
Heat Map
Color
Blue-Green-Yellow
Range:
Merge
Merge by k-means
Clusters:
50
Clustering
Rows:
Clustering (opt. ordering)
Columns:
Clustering (opt. ordering)
Split By
Rows:
function
Columns:
(None)
Annotation & Legends
Show legend
Stripes with averages
Row Annotations
Text
(None)
Color
(None)
Send Automatically
-0.549
0.566
Proteas
Resp
Ribo

11. COMPLETE MACHINE LEARNING WORKFLOW IN ORANGE

A complete workflow in Orange typically follows these steps:

1. Load dataset
2. Explore and understand structure
3. Preprocess data
4. Select features
5. Train model
6. Evaluate model
7. Visualize results
8. Predict new data

The workflow structure helps users understand the logical sequence of machine learning steps. Since everything is visible on the Canvas, it improves conceptual clarity.

12. CONCLUSION

Orange Data Mining provides a powerful yet simple platform for performing machine learning tasks without writing complex code. Its visual workflow approach helps users clearly understand each stage of the process, from data loading and preprocessing to model evaluation and prediction.

With strong visualization capabilities, multiple algorithm support, and an interactive interface, Orange is highly suitable for:

- Academic learning
- Research experiments
- Practical data analysis projects
- Beginner-level machine learning training

Overall, Orange bridges the gap between theoretical machine learning concepts and practical implementation in an intuitive and structured manner.

13. REFERNCES:

Official Website (Primary Source)

Orange Data Mining. (n.d.). *Official website*. Retrieved from: https://orangedatamining.com/

Official Documentation

Orange Data Mining. (n.d.). *Documentation and user guide*. Retrieved from: https://orange3.readthedocs.io/

Research Paper (Highly Recommended for Academic Submission)